

Adapting Clinical Chemistry Plasma as a Source for Liquid Biopsies

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Spencer C Ding, Jingru Yu, Tiepeng Liao, Lauren S Ahmann, Yvette Y Yao, Chandler Ho, Linlin Wang, Benjamin A Pinsky, Wei Gu 

Department of Pathology, School of Medicine, Stanford University, Stanford, United States • Department of Molecular and Medical Genetics, Oregon Health & Science University, Portland, United States • School of Medicine, University of British Columbia, Vancouver, Canada • Clinical Laboratories, Stanford Health Care, Stanford, United States • Department of Laboratory Medicine, School of Medicine, University of California San Francisco, San Francisco, United States • Department of Medicine, Division of Infectious Diseases and Geographic Medicine, School of Medicine, Stanford University, Stanford, United States

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eLife Assessment

This **important** work provides a new method to extract cfDNA from residual plasma from heparin separators for molecular testing. The evidence supporting the authors' claims is **convincing**, although some further metrics should also be evaluated. This finding will be interesting to people working in epigenomics and infectious disease diagnostics.

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Abstract

Background

Circulating cell-free DNA (cfDNA) has become a valuable analyte for molecular testing but requires specialized collection tubes or immediate processing. We investigated the feasibility of using residual plasma from heparin separators, which are routinely used in clinical chemistry, as an accessible and underutilized source for cfDNA testing.

Methods

We analyzed matched plasma samples collected in EDTA, Streck, and heparin separators in a Healthy Cohort (n = 5) and matched samples collected in EDTA and heparin separators from a Hospital Cohort derived from viral PCR-positive patients (n = 34). Whole-genome sequencing and genome-wide enriched methylation sequencing were performed to evaluate concordance across multiple benchmarks, including metagenomics, chromosomal copy number, methylome, and fragmentomics.

Results

In the Healthy Cohort, methylation patterns were correlated (Pearson's $r = 0.92$ – 0.93) between tube types, and fragmentation features were preserved with a modal size peak at 166 bp and a consistent top 10 end motif ranking across tube types ($n = 5$). In the Hospital Cohort, heparin separators showed a strong concordance with matched EDTA tubes for viral detection ($n = 34$, Pearson's $r = 0.99$), copy number alteration profiling ($n = 6$, Pearson's $r = 0.86$ – 0.98), and methylation patterns ($n = 12$, $r = 0.83$ – 0.93).

Conclusions

Residual plasma from routine clinical chemistry tests can provide a vast, untapped resource for cfDNA analysis.

Background

Plasma cell-free DNA (cfDNA) is widely used in clinical testing(1 [↗](#)), including non-invasive prenatal testing (NIPT)(2 [↗](#)–5 [↗](#)), cancer testing(6 [↗](#)–8 [↗](#)), infectious disease testing(9 [↗](#), 10 [↗](#)), and organ transplantation monitoring(11 [↗](#), 12 [↗](#)). In cancer testing, cfDNA is actively studied for the detection of residual cancer disease(13 [↗](#)) and early detection (14 [↗](#), 15 [↗](#)).

Reliable cfDNA analysis typically depends on specialized preservative tubes (e.g., Streck) or rapid processing of EDTA tubes(16 [↗](#)). However, both methods limit accessibility by requiring specific tubes or on-call personnel, which adds costs and restricts cfDNA testing in resource-limited settings.

In contrast, heparin separators are standardly used in clinical settings, ubiquitous in everyday phlebotomy trays at many institutions for routine chemistry testing, such as the basic metabolic panel. For example, at our institution alone (Stanford Health Care), over 900,000 of these tubes are processed annually. These tubes contain a gel barrier that separates plasma from cellular components following routine centrifugation. They are typically handled through high-throughput, automated, and CLIA-compliant workflows that incorporate prompt centrifugation and refrigerated storage.

This study explores the potential utilization of millions of leftover plasma samples and the associated healthcare infrastructure as an untapped resource for cfDNA research and clinical testing. To assess the feasibility of using heparin separators for cfDNA testing, we conducted whole-genome sequencing (WGS) and the FLEXseq methylation assay(17 [↗](#)) on paired samples collected from the same venipuncture in both heparin separators and gold standard EDTA (and/or Streck) tubes. Our evaluation spanned multiple benchmarks relevant to a broad assortment of cfDNA tests, including metagenomics(9 [↗](#), 18 [↗](#)), methylation profiling (17 [↗](#), 19 [↗](#)), copy number alterations (CNAs)(20 [↗](#)), epigenetic tissue-of-origin mapping(19 [↗](#)), and cfDNA fragmentomics(21 [↗](#)).

Results

Heparin separators preserve cfDNA integrity under ideal conditions

To evaluate the potential of repurposing leftover plasma from heparin separators, we conducted a controlled comparison in the Healthy Cohort ($n = 5$) using matched samples collected from the same blood draw in heparin separators and in gold-standard EDTA and Streck tubes (**Fig. 1a**, Table S1).

Fragment size distributions were highly similar across all tube types, with a modal peak at ~166 bp, consistent with known cfDNA profiles (**Fig. 1b**)(4,24). End motif analysis of 256 4-mers showed comparable rankings (**Fig. 1c**), suggesting that heparin separators do not perturb fragmentation patterns when processed immediately.

To assess the preservation of epigenomic features, we calculated methylation levels at individual CpG sites using the FLEXseq methylation assay's output. CpG sites with $\geq 30\times$ coverage showed strong correlations between heparin separator and EDTA (Pearson's $r = 0.93$, **Fig. 1d**, left), and between heparin separator and Streck ($r = 0.93$, **Fig. 1d**, right). Tissue deconvolution revealed highly concordant cell type proportions across all collection tube types (**Fig. 1e**, $P = 0.86$, PERMANOVA), with consistent results across all five cases (Heparin separator versus EDTA: Pearson's $r = 0.92$ - 0.93 ; heparin separator versus Streck: Pearson's $r = 0.92$ - 0.93 ; **Supplemental Fig. 1**). These findings show that heparin separators preserve the biological status of methylation under ideal processing conditions.

To mimic clinical storage, we tested the impact of delayed processing. Blood from a healthy volunteer was collected into EDTA, heparin separator, and non-gel heparin tubes, then divided into two sets (**Supplementary Fig. 2a**). One set was processed immediately, while the other was refrigerated (4°C) for one hour before the first spin and an additional four days before the second spin. Fragment size profiles were identical (**Supplementary Fig. 2b**), confirming that refrigerated storage does not compromise cfDNA integrity.

Hospital Cohort: Residual plasma from Heparin separators after clinical usage

To evaluate the feasibility of heparin separator tubes for cfDNA analysis, we identified 34 matched pairs of viral PCR-positive plasma samples (viral load >1000 IU/mL) collected in both heparin separator and EDTA tubes from the same blood draw. These cases were retrieved by querying the electronic medical record systems at Stanford ($n=28$) and UCSF ($n=6$) for samples that tested positive by the Clinical Virology Laboratories. All samples underwent cfDNA extraction, followed by whole-genome sequencing (WGS) to evaluate viral load, copy number alterations (CNAs), and fragmentation profiles, as well as methylation profiling via FLEXseq.

Leftover plasma from the heparin separators enables viral load detection

We performed metagenomic sequencing to compare viral load measurements and assess concordance between plasma collected in EDTA and heparin separators (**Fig. 2**). WGS achieved a median coverage of $4.52\times$ (IQR: 1.76 - $5.08\times$). Viral read counts normalized to the number of reads aligned to the human genome were highly correlated between plasma collection tube types

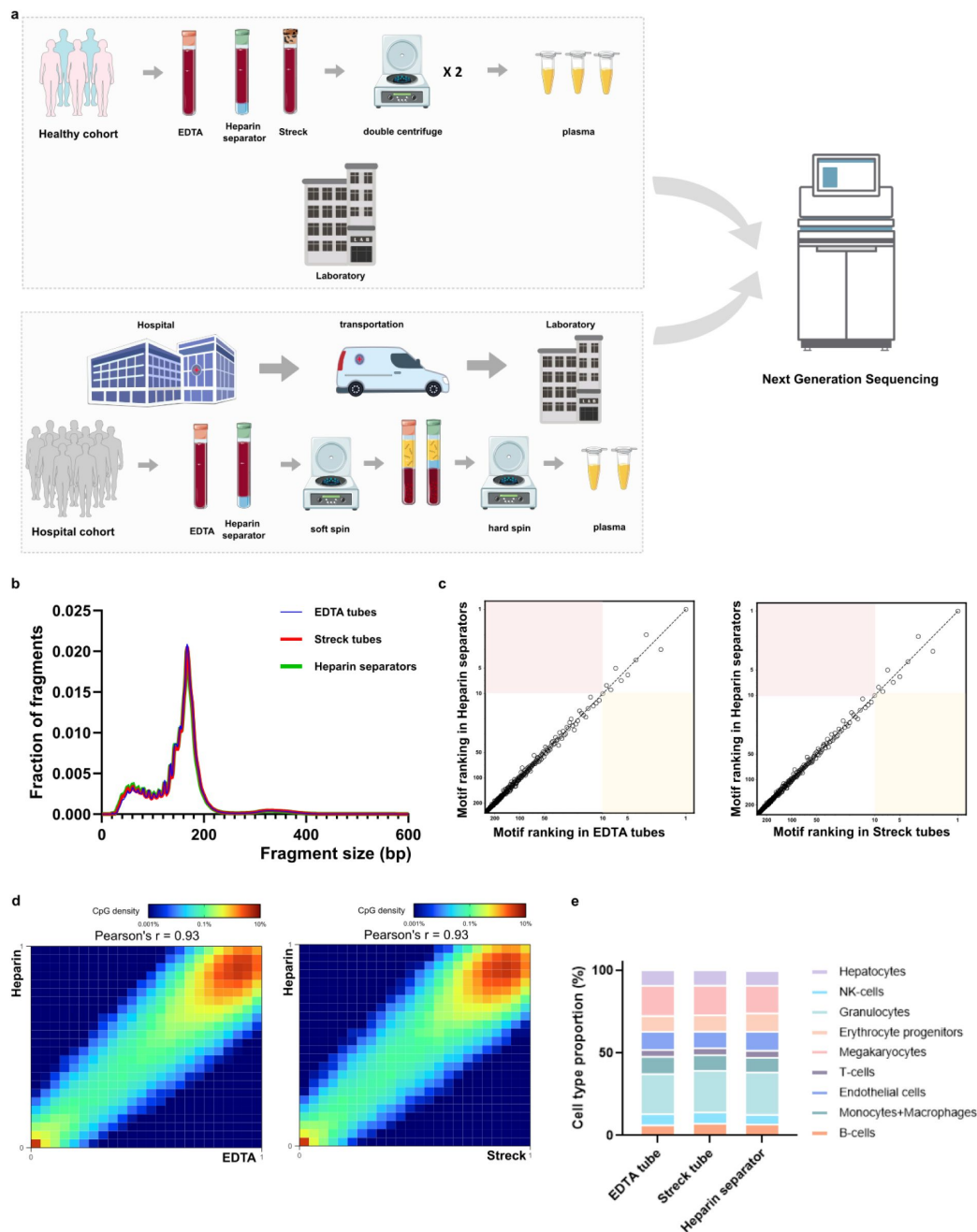


Figure 1.

Overall schematic and Healthy Cohort's comparative performance

a. Schematic illustration of the experimental design across two cohorts. The Healthy Cohort was a controlled experiment where plasma samples from healthy volunteer donors were collected freshly in EDTA, Streck, and heparin separators. The Hospital Cohort plasma samples were retrospectively collected in EDTA and heparin separators as part of patient care and leftover from routine clinical testing. The Hospital Cohort samples underwent an initial soft spin during the routine clinical workflow and then a hard spin after refrigerated storage. Processed samples were analyzed using whole-genome sequencing (WGS) and/or FLEXseq methylation testing. b. Healthy Cohort fragment size distribution of cfDNA collected across the three tubes. c. Healthy Cohort comparison of end motif rankings in heparin separator samples versus EDTA or Streck tubes. d. Healthy Cohort heatmap and correlation of methylation beta values in paired samples from a healthy donor (P140). Heparin separators (y-axis) are compared to EDTA and Streck tubes (x-axis). e. Healthy Cohort estimation of cell-type proportions using methylation deconvolution for each collection tube.

(Pearson's $r = 0.99$, $P < 0.001$), with no significant difference observed (Wilcoxon signed-rank test, $P = 0.27$). This high level of concordance was consistently observed across institutions and among distinct viral types.

Leftover plasma from the heparin separator enables copy number profiling

CNAs are commonly observed in cancer and can be detected in cfDNA as a surrogate marker for tumor burden. To evaluate whether heparin separators preserve CNA signals, we compared the genome-wide copy number profiles of samples from heparin separators and EDTA tubes. CNA profiles were generated using a 500 kbp bin size, with a median of 97.68 million reads per sample (range: 0.73–360.73 million). Representative genome-wide CNA plots from a matched pair (P171: heparin separator; P187: EDTA) revealed similar patterns of chromosomal gains and losses (**Fig. 3a**). Genome-wide $\log_2$ copy ratio deviations were concordant between tube types (Pearson's $r = 0.97$; **Fig. 3b**). Similar patterns and concordance were observed in other CNA-positive samples ($n = 5$, Pearson's $r = 0.86$ – 0.99 ; **Supplementary Fig. 3**). Tumor fraction estimates based on copy number also correlated strongly between matched samples (Pearson's $r = 0.97$; **Fig. 3c**). These findings demonstrate that plasma from heparin separators is suitable for CNA detection and tumor fraction estimation.

Correlated Methylation Profiling from Heparin Separator Plasma

To assess the reliability of methylation profiling across blood collection tube types, we analyzed the available residual matched plasma samples in the Hospital Cohort ($n=12$). Using FLEXseq(17), a high-resolution methylation assay, we observed strong genome-wide concordance of CpG methylation levels between heparin separator and EDTA tubes (P168: heparin separator, P185: EDTA tube; Pearson's $r = 0.90$; **Fig. 4a**; other cases in **Supplementary Fig. 4**). To evaluate the biological relevance of the cfDNA methylation profiles, we performed cell type deconvolution on the paired plasma(19). The results showed comparable tissue-of-origin proportions between plasma collected in heparin separator and EDTA tubes (**Fig. 4b**). These findings demonstrate that cfDNA methylation signatures are well preserved in heparin separator plasma and are reliable for epigenomic applications such as cell-of-origin estimation.

Comparable cfDNA yield deduced from the sequencing result

We quantified cfDNA concentrations (normalized to plasma volume) in the 25 matched pairs from the Healthy Cohort and the Hospital Cohort with sufficient leftover DNA. Qubit measurements revealed variability across samples (0.14–43.19 ng/mL; **Fig. 5a** & **Fig. 5b**) and poor correlation between EDTA and heparin separator data. However, Qubit measurements are related to both addressable (intact short fragments) and non-addressable (genomic length or broken) DNA, whereas NGS output is related to only addressable DNA.

To enable precise measurements of the addressable DNA, we spiked in lambda DNA at a constant quantity across all samples at the beginning of the library preparation and normalized human-aligned read counts to this internal control. This strategy produces a measure (Human/Lambda ratio) proportional to the addressable DNA content of the sample. This normalized ratio yielded strong concordance between EDTA and heparin separator samples ($R^2 = 0.998$; **Fig. 5c**), indicating that library preparation effectively mitigated preanalytical variability.

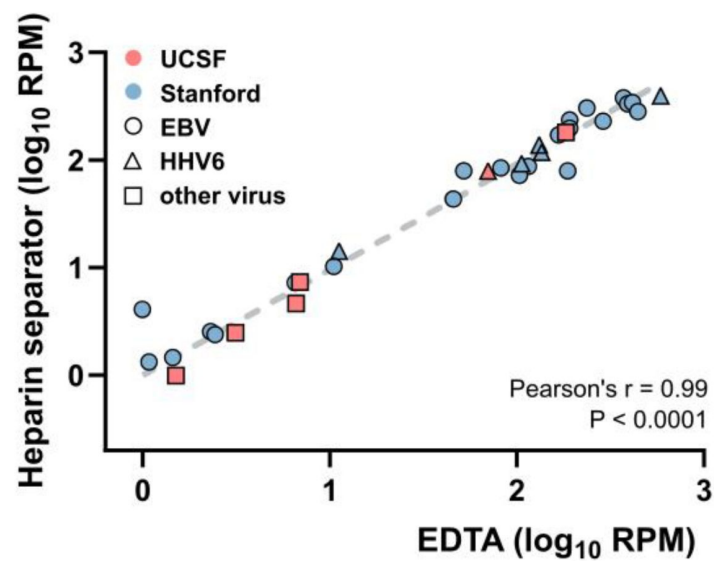


Figure 2.

Viral load correlation of the Hospital Cohort

Correlation of plasma viral DNA reads detected by whole genome sequencing in paired cases collected in EDTA and heparin separators. Samples were collected at two institutions (Stanford - blue, UCSF - red, EBV - circle, HHV6 - triangle, other virus - square). RPM: reads per million human-aligned reads.

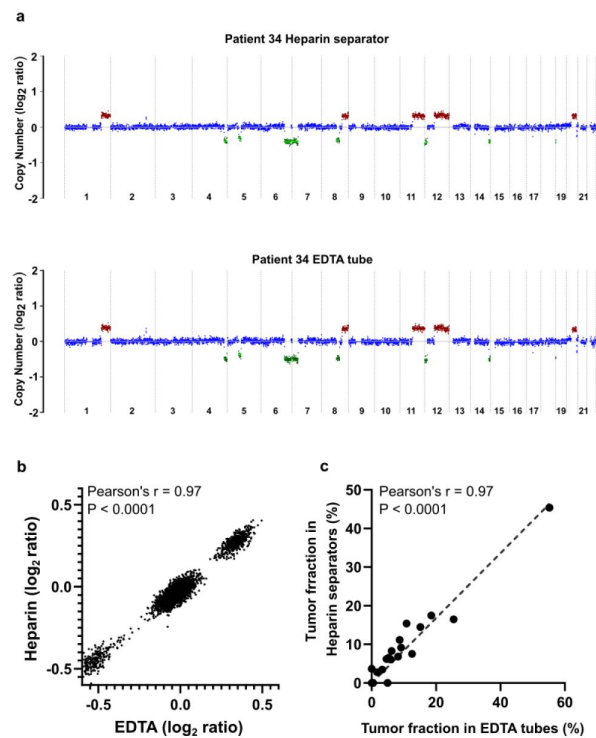


Figure 3.

Copy number analysis of the Hospital Cohort

a. An example of genome-wide copy number profiles from Patient 34. Case P171 was collected in a heparin separator; Case P187 was collected in EDTA tube. b. Comparison of $\log_2$ copy ratios between P171 and P187 based on 500 kb bin sizes. c. Comparison of tumor fraction in paired cases ($n=28$ pairs) inferred from copy number analysis.

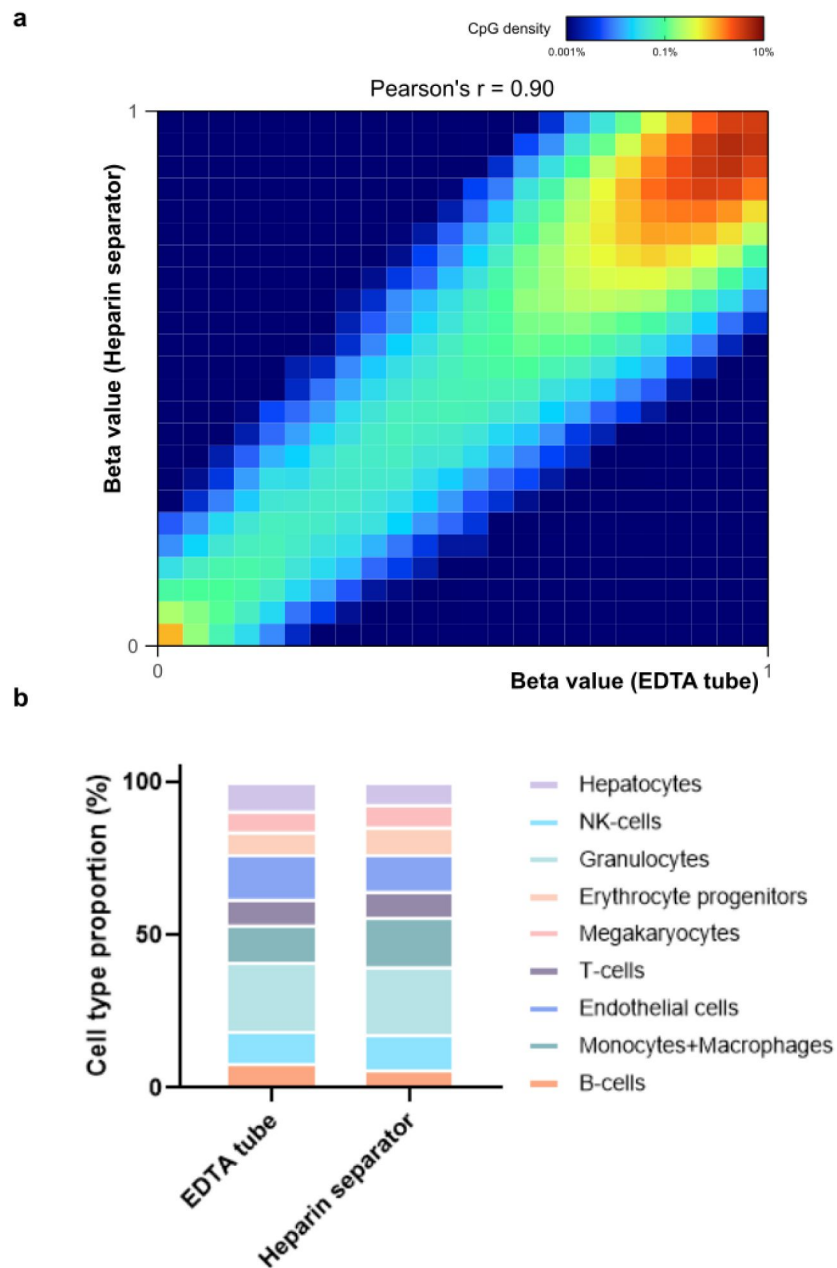


Figure 4.

Methylation correlation and cell type deconvolution analysis in the Hospital Cohort

a. Heatmap showing methylation beta values in paired samples (Patient 31, P185 versus P168). b. Estimated cell-type proportions in the paired samples (P185 versus P168).

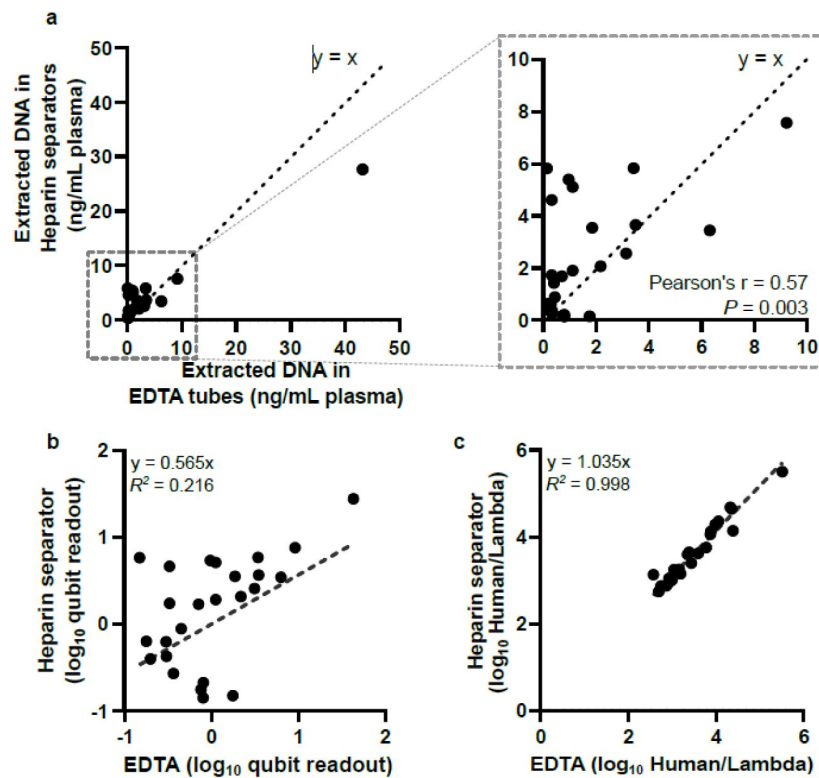


Figure 5.

Correlation of DNA quantity

a. Scatter plot of the normalized extracted DNA concentration based on Qubit reading. Data is from 20 individuals of the Hospital Cohort and 5 individuals from the Healthy Cohort using heparin separators and EDTA tubes. Normalized DNA concentration (ng/mL plasma) in heparin separator samples (y-axis) vs. EDTA tube samples (x-axis), quantified by dsDNA Qubit and normalized to the input volume of plasma. b. Scatter plot of the log-scale normalized concentration (Qubit readout) in the heparin separators (y-axis) vs. the EDTA tube samples (x-axis). c. Scatter plot of the ratio of human reads to a constant lambda phage DNA spike-in detected in the heparin separator sample (y-axis) vs. the EDTA tube samples. This spike-in measurement is proportional to the addressable DNA quantity of the sample.

Leftover plasma from the heparin separator alters fragmentation profiles

Fragmentation features of cfDNA, such as size distribution and end motifs, have been linked to nuclease activity and are known to reflect host physiology and pathology(25 [↗](#)–27 [↗](#)). To evaluate whether heparin separators preserve these fragmentation features, we examined two well-characterized fragmentomic features: size distribution and end-motif. Both EDTA and heparin separator plasma samples exhibited expected peak sizes near 70 bp and 170 bp (**Supplementary Fig. 5** [↗](#)). However, heparin separator plasma showed a modest increase in short fragments and a slightly diminished ~166 bp peak, indicating a shorter size distribution. Analysis of 4-mer motifs at the 5' ends of cfDNA fragments revealed significant differences in four of the six most frequent motifs (i.e., CCCA, CCAG, CCTG, and CAAA; $P < 0.05$, paired Student's t-test), suggesting disrupted cleavage preference. These altered fragmentation profiles in the Hospital Cohort suggest that plasma samples collected in heparin separators may have influenced cfDNA fragmentation signatures. Given that the fragmentation profiles were concordant in the Healthy Cohort, the differences seen here may be due to degradation during delays in clinical lab processing after blood collection and prior to refrigerated storage.

Discussion

Our study explores the untapped potential of repurposing residual plasma from heparin separators as a practical and underutilized resource for circulating cfDNA-based research and diagnostic applications. Using matched samples from a Healthy Cohort and a real-world Hospital Cohort, we demonstrate that the leftover plasma is suitable for metagenomic viral detection, copy number profiling, and genome-wide methylation analysis, with performance comparable to gold-standard collection methods.

Although heparin separators have been previously used in COVID-19 cfDNA studies(28 [↗](#),29 [↗](#)), their performance has not been systematically evaluated against standard collection methods. Historically, concerns about heparin-induced PCR inhibition and genomic DNA leakage from cells have limited the use of heparin tubes, although the available evidence remains inconclusive. For instance, Lo et al. reported no significant difference in DNA concentration between EDTA and heparin tubes when plasma was processed promptly within 2 hours after blood collection, suggesting minimal gDNA release(30 [↗](#)). In contrast, Gerber et al. reported elevated cfDNA levels in heparin tubes compared to specialized tubes such as PAXgene, Roche cfDNA, and Streck, even at time point 0(31 [↗](#)). The results in our Healthy Cohort suggest no significant differences in samples immediately processed or refrigerated at 4°C. Unlike earlier studies that relied on PCR-based assays and used standard heparin tubes, our study utilized heparin separators with a gel barrier to physically separate plasma from cells. Moreover, modern cfDNA sequencing workflows include multiple purification steps that could effectively reduce heparin interference on PCR and preferentially retain short cfDNA fragments, minimizing the impact of gDNA contamination on downstream analysis.

Fragmentation profiles were consistent across tube types in the Healthy Cohort but variable in the Hospital Cohort, likely reflecting preanalytical differences before laboratory processing. Prior research has shown that incubating blood with heparin at room temperature for 6 hours alters cfDNA fragmentation profiles, possibly due to heparin-induced nucleosomal disruption and enhanced DNase I activity(32 [↗](#)). In contrast, prompt refrigeration in the Healthy Cohort preserved cfDNA integrity, potentially by inhibiting nuclease activity.

Our study demonstrates the feasibility of repurposing residual plasma from heparin separators for sequencing-based cfDNA analysis, offering a collection method that integrates seamlessly into existing clinical workflows. By utilizing samples already collected during routine care, this approach eliminates the need for additional venipuncture and enables retrospective access to clinically important cases, including those where informed consent could not be obtained at the time of collection. While promising, the approach is constrained by the limited residual plasma volume (0.5–1.5 mL), which may not meet the input requirements of applications such as minimal residual disease detection. Moreover, variability in transport and storage conditions prior to processing, as observed in the Hospital Cohort, can affect cfDNA integrity. The observed alteration in fragmentation profiles highlights the need for cold-chain transportation protocols to minimize variability and maintain cfDNA quality. Future studies should also validate this approach in larger, clinically diverse populations and assess its integration into routine preanalytical workflows.

In summary, residual plasma from heparin separators can serve as a viable source for cfDNA analysis, with stable results for copy number variation, viral load, and methylation profiling despite fragmentomic sensitivity to preanalytical variability. Repurposing residual plasma from routine blood tests could greatly simplify and expand cfDNA accessibility across a diverse range of patients.

Methods

Study design and subject selection

Healthy Cohort: Five healthy adult volunteers (28–40 years old) donated blood with informed consent as approved under Stanford IRB (71230). Whole blood was collected from each donor via a single venipuncture into three tube types: EDTA, Streck, and heparin separators.

Hospital Cohort: Clinical samples were from two institutions, Stanford and UCSF. At Stanford, 28 matched pairs of remnant samples were obtained (collected between 2022 and 2025) under a no-patient-contact protocol (IRB 58461). Patients gave written consent for the use of residual clinical samples. Patients positive for Epstein-Barr virus (EBV), human herpesvirus 6 (HHV6), BK virus, adenovirus, or Torque Teno Virus were identified based on routine clinical viral PCR testing performed by the Stanford or UCSF Clinical Virology Laboratories. These viral tests were typically conducted in the context of post-transplant surveillance or infectious disease diagnostics. Residual EDTA plasma samples were processed by the clinical laboratory within four hours of collection. Matched heparin separator plasma specimens drawn at the same time were retrieved from the Stanford Clinical Chemistry Laboratory. Only matched plasma pairs with volumes exceeding 0.5 mL were included in the study. Additional sample pairs from UCSF ($n = 6$) were obtained (collected between 2017 and 2020) under a similar no-patient-contact IRB protocol (CA-0161925).

Sample collection and processing

Hospital Cohort: Upon arrival, plasma samples in EDTA tubes and heparin separators were immediately centrifuged at 1300 g for 10 minutes using a Cobas 8100 automated workflow. Following completion of routine clinical testing, the residual plasma is stored and transported at approximately 4°C for up to 8 days before further processing. EDTA plasma was separated into a different container, and heparin separator plasma was kept in the same original collection tube, but physically isolated above the gel separator. These procedures were conducted by the hospital staff as part of routine clinical care in accordance with a CLIA-compliant standard operating procedure. The plasma supernatant was identified by a research team, deidentified, transferred to our research lab at refrigeration, and subjected to a second centrifugation at $16,000 \times g$ for 10 minutes. Plasma aliquots from both EDTA tubes and heparin separators were stored at -80°C until downstream analysis. UCSF cases were processed as previously described⁽¹⁸⁾.

DNA extraction

Cell-free DNA (cfDNA) was extracted from 0.5 to 1 mL of plasma using the Maxwell RSC 48 instrument (Promega) with the Maxwell RSC cfDNA Plasma Kit (AS1480), following the manufacturer's instructions. Samples were processed in parallel in batches. cfDNA concentrations were quantified using the Varioskan spectrophotometer (Thermo Fisher Scientific). UCSF cases were extracted as previously described([18](#) .

Whole genome sequencing DNA library preparation

A median of 18.65 ng of extracted cfDNA (range: 3.00-51.70 ng) and 2.5 ng of fragmented Lambda DNA (NEB #N3013) as internal control was used for whole-genome sequencing (WGS) library preparation with the NEBNext Ultra II DNA Library Prep Kit (New England Biolabs, #E7645L), following the manufacturer's protocol. PCR primers were NEBNext Multiplex Oligos for Illumina (NEB #E6444L). Adapter-ligated DNA was amplified for 20 cycles using the Q5 master mix, following the manufacturer's instructions. UCSF cases were processed as previously described([18](#) .

FLEXseq DNA library preparation

FLEXseq DNA libraries were prepared as previously described. Briefly, extracted cfDNA molecules were adapter-ligated, dephosphorylated, digested with MspI (NEB, #R0106T), adapter-ligated a second time with custom adapters, subjected to NEBNext Enzymatic Methyl-seq Conversion Module (New England Biolabs, #E7125L), and PCR amplified.

Sequencing and alignment

Paired-end sequencing was performed on the NovaSeq X platform (Illumina) at the University of Chicago Genomics Core Facility using the 25B Reagent Kit (300 cycles). For WGS data, sequencing adapters were trimmed using Cutadapt v3.5 with the following parameters: -a AGATCGGAAGAGC -A AGATCGGAAGAGC --minimum-length 25. Trimmed reads were aligned to the hg38 human reference genome, as well as EBV, HHV6, and other viral genomes using BWA v0.7.17. PCR duplicates were removed using Samtools.

FLEXseq data were processed following the protocol described by Yu et al.([17](#) .

Briefly, reads were trimmed to remove adapters and filtered based on quality scores. High-quality reads were aligned to the hg38 genome using Bismark v0.23.0. Unconverted and duplicated reads were removed using Bismark. All SNP-containing regions were masked for de-identification using Bedtools.

Viral load analysis

The ratio of reads aligned to viral genomes (e.g., EBV, HHV6) relative to every one million reads aligned to the human genome (hg38) was calculated and log-transformed. Pearson correlation analysis was performed between matched tube types, with the linear regression constrained to pass through the origin (0, 0).

CNA analysis

Copy number alterations were analyzed using ichorCNA on autosomal chromosomes([22](#) .

HMMcopy's readCounter was used to generate 500 kb bin-level Wig files from aligned BAM files, which included only reads with a mapping quality score >20. Reference files for GC content (gcWig), mappability score (mapWig), and centromere positions were obtained from the ichorCNA

GitHub repository corresponding to the selected bin size. Panel-of-normal (PoN) Wig files were generated separately using the cases in the Healthy Cohort in the EDTA and heparin separator groups, respectively.

Tumor purity was estimated based on log₂ ratio outputs from ichorCNA using previously reported calculations⁽²³⁾.

$$TumorPurity(\%) = \frac{1 - 2^{(log_2 \text{ ratio})}}{1 - \frac{(\text{assumed ploidy})}{2}} \times 100$$

Methylation correlation analysis

Methylation calls were extracted using the `bismark_methylation_extractor`. For each CpG site, the methylation level was calculated as the ratio of methylated reads to total coverage depth. CpG sites with coverage $\geq 30\times$ were retained for correlation analysis between matched samples.

Tissue-of-origin analysis

Tissue-of-origin deconvolution was performed using cell type-specific unmethylated markers as previously described by Loyfer et al.⁽¹⁹⁾. The top 250 unmethylated CpG markers for each cell type were extracted from the reference dataset. Markers located on sex chromosomes or overlapping with common single nucleotide polymorphisms (SNPs) were excluded. Deconvolution was conducted using the `uxm` function based on a panel of nine reference cell types: B cells, T cells, natural killer (NK) cells, granulocytes, monocytes/macrophages, endothelial cells, erythroid progenitors, megakaryocytes, and hepatocytes.

cfDNA fragmentomics

Fragment size distributions were generated for each sample by calculating the frequency across fragment lengths between 50 and 600 bp. End motif frequencies were also calculated for each sample. The Shapiro-Wilk test was performed separately for each motif in different tubes (EDTA, Heparin separator, and Streck) to assess whether the frequency distributions followed a normal distribution. For the Healthy Cohort, differences among the three collection tubes were evaluated using the Friedman test. For the Hospital Cohort, statistical significance between tube types was assessed using the Paired Student's t-test. To control the false discovery rate (FDR), multiple testing corrections were applied using the Benjamini-Hochberg procedure across all tested motifs. Significance was determined based on FDR-adjusted *P*-values.

Data Availability

Raw WGS and genome-wide enriched methylation sequencing data are available for the Healthy Cohort (*n* = 5) with informed consent. FASTQ files were uploaded to the Sequence Read Archive (SRA), with the Bioproject ID of PRJNA1260066. Viral reads detailed statistic files (*n*=28), methylation calls (*n*=17), copy number plots (*n*=6), and fragment length files (*n*=34) are available on Zenodo (<https://doi.org/10.5281/zenodo.15367094>).

Acknowledgements

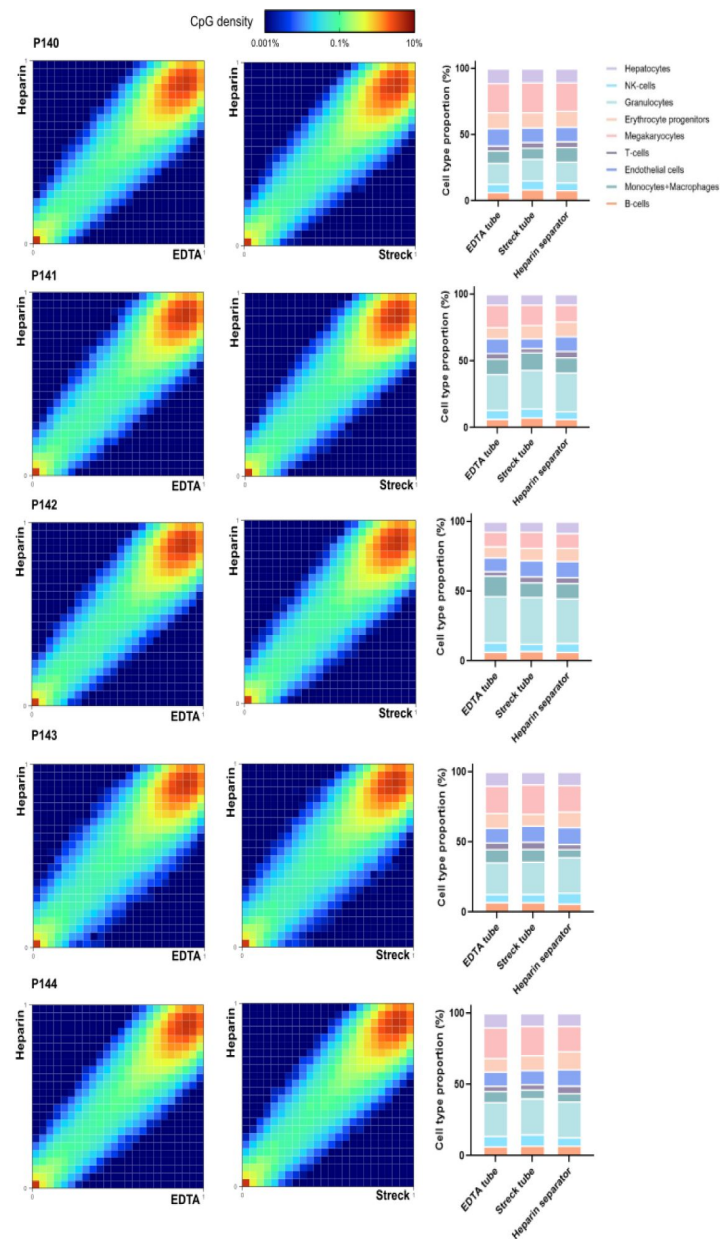
This work was funded by an NIH K08 (CA230156) grant, a Burroughs-Wellcome CAMS Award to WG. We thank the members of the Stanford Clinical Molecular Genetic Pathology, Clinical Virology, and Clinical Chemistry Laboratories, and UCSF Clinical Virology and Clinical Chemistry

Laboratories for saving residual specimens. We thank Debbie Chan and Dianna Ng for their critical insights at the inception of this project and Ruben Luo for facilitating Clinical Chemistry collections at Stanford. We thank the University of Chicago Genomics Facility (*RRID:SCR_019196* [↗](#)) for Novaseq sequencing.

Additional files

Supplementary table [↗](#)

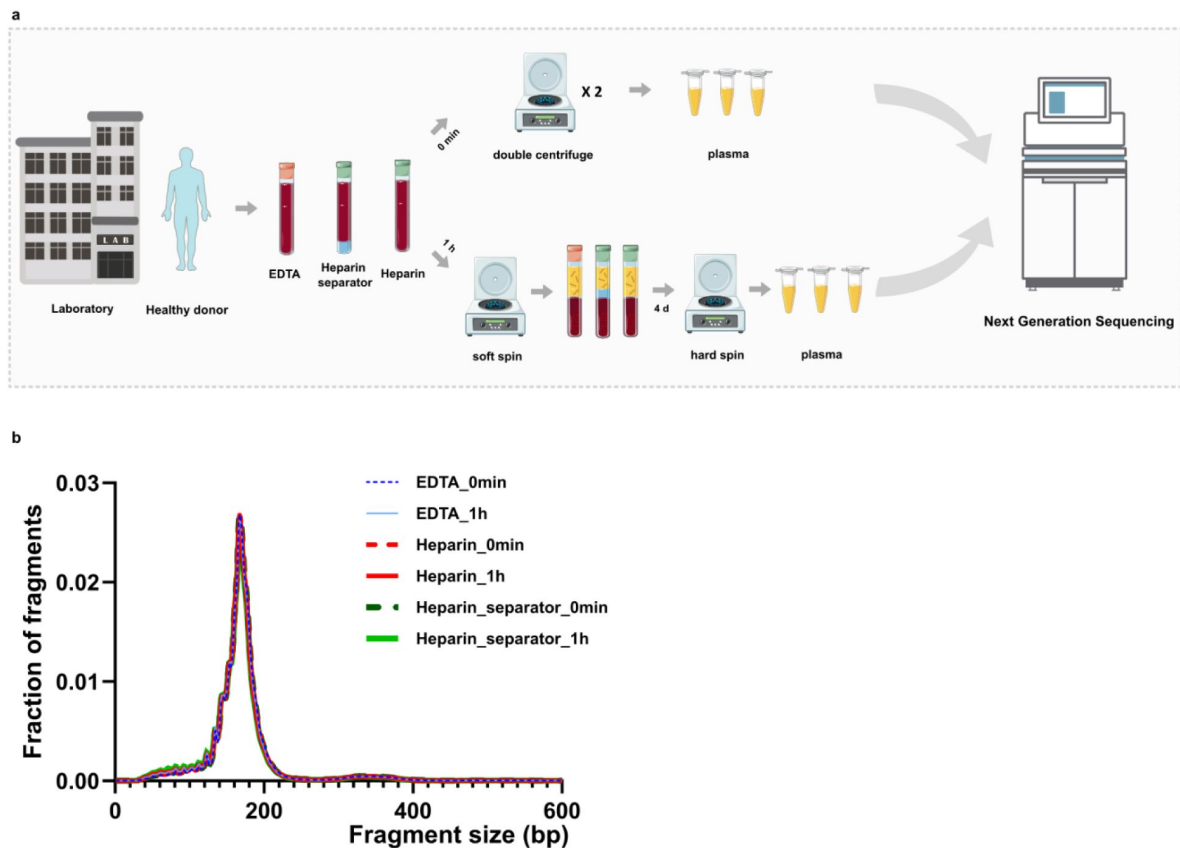
Supplementary Figures



Supplementary Figure 1.

Healthy Cohort methylation correlation and cell type deconvolution

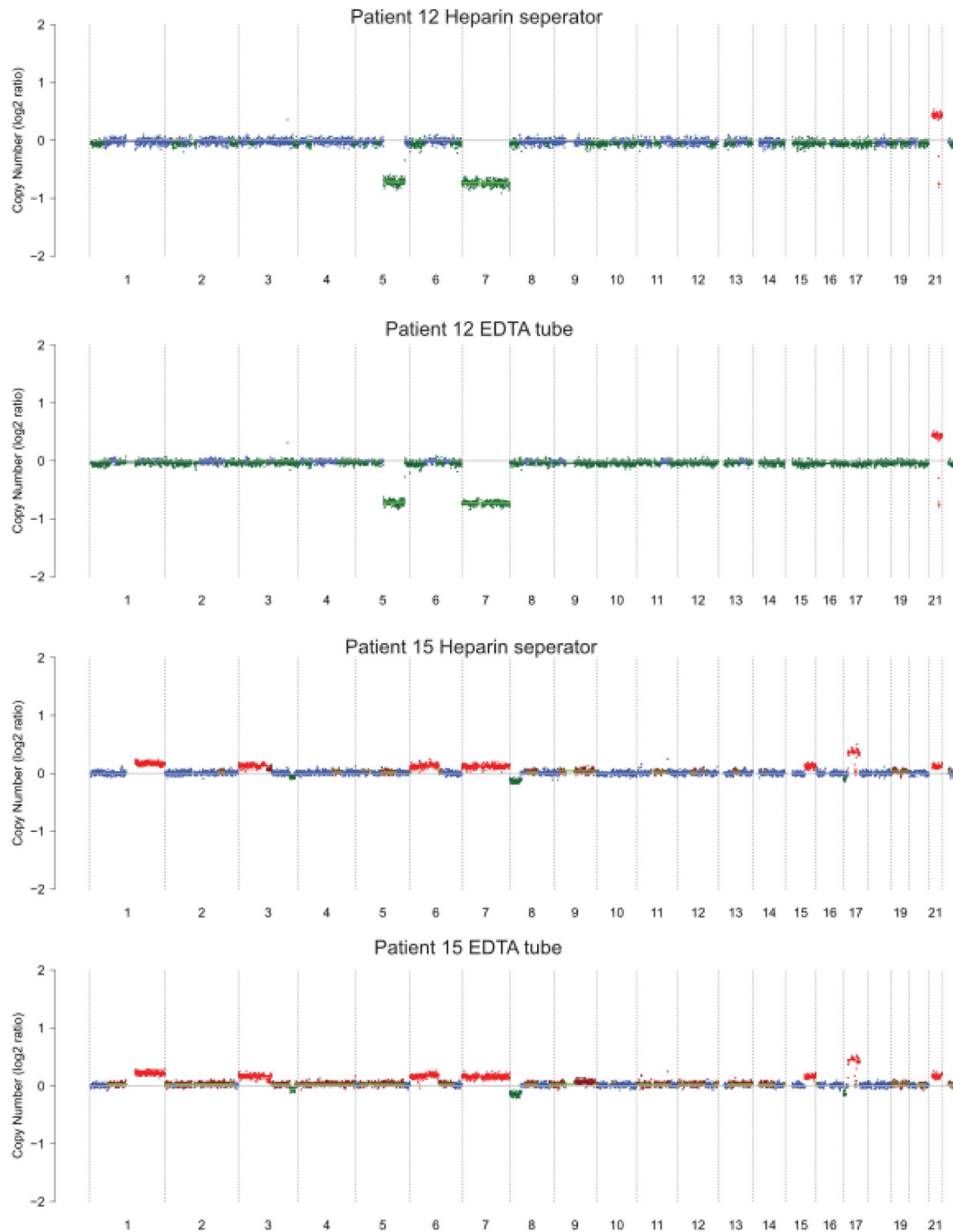
Heatmap and correlation of methylation beta values with cell type deconvolution results in paired samples from five healthy donors.



Supplementary Figure 2.

Healthy Cohort size distribution of cfDNA in EDTA and heparin tubes over time mimicking the clinical workflow

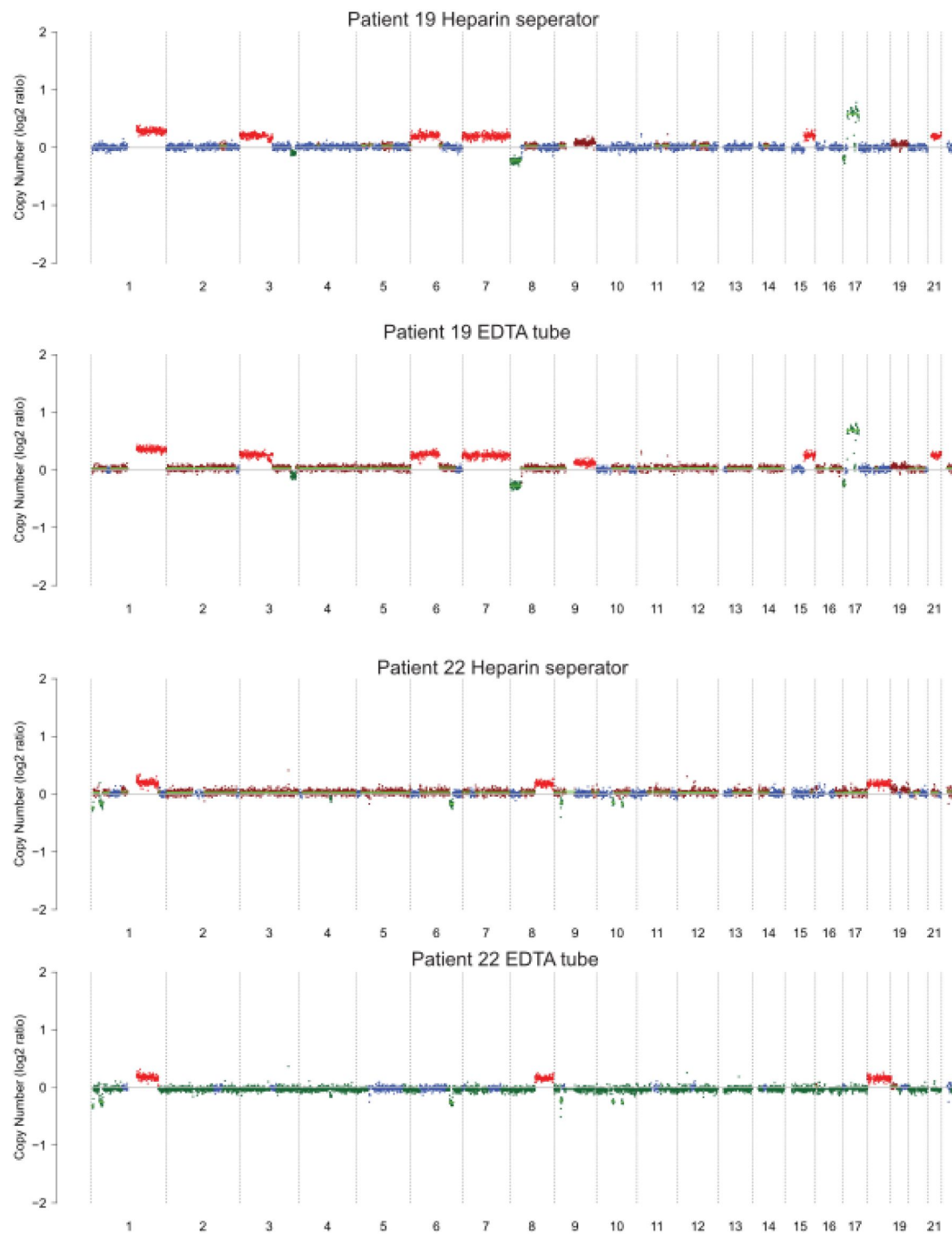
a) Schematic of the time-lapsed controlled experiment mimicking the clinical workflow. b) The fragment size distribution of cfDNA across all conditions. "0 min" represents immediate centrifugation, while "1 h" indicates samples stored at 4°C for 1 hour before the soft spin and for 4 days at 4°C before the hard spin.



Supplementary Figure 3.

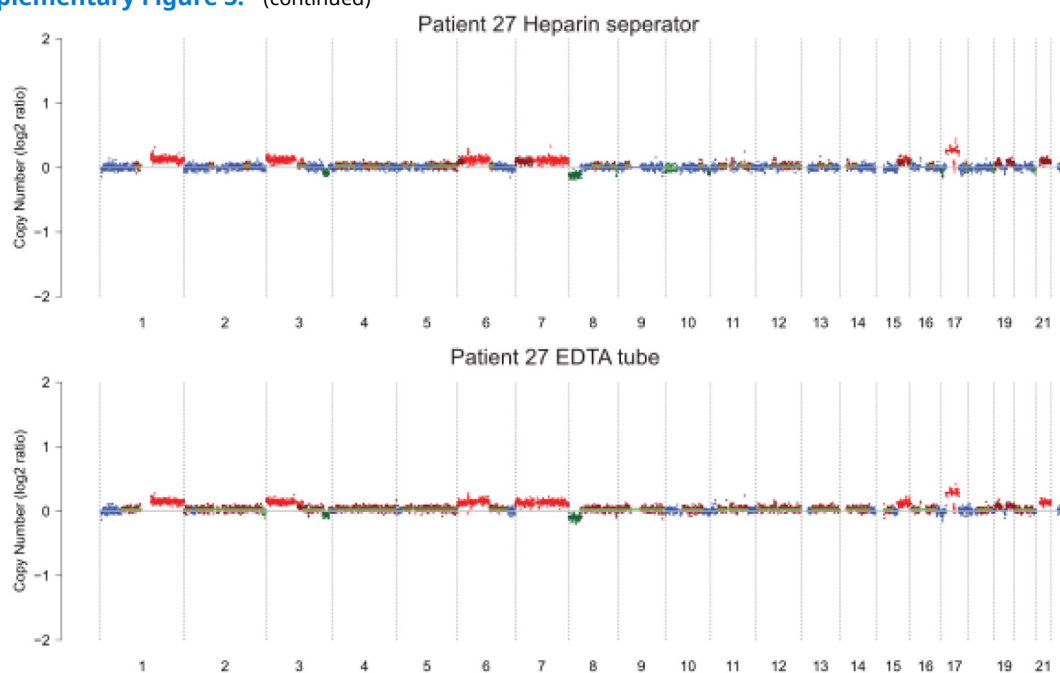
Copy number analysis across Hospital Cohort cases

Comparison of other CNA-positive profiles of paired cases in the Hospital Cohort (n=5).



Supplementary Figure 3. (continued)

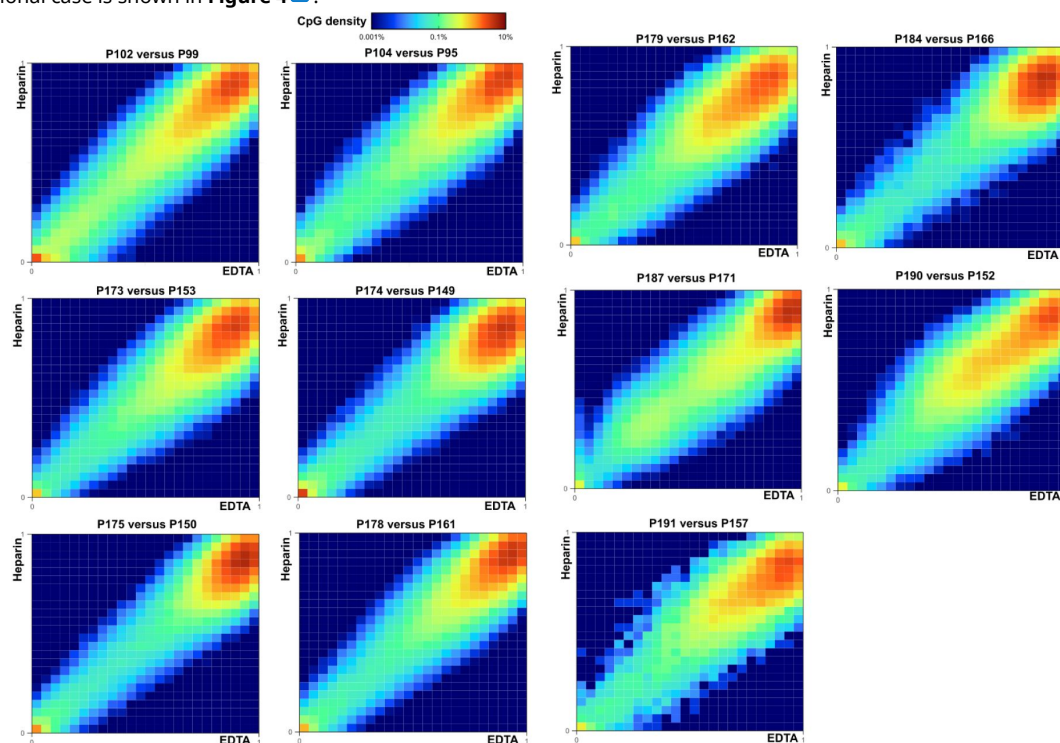
Supplementary Figure 3. (continued)

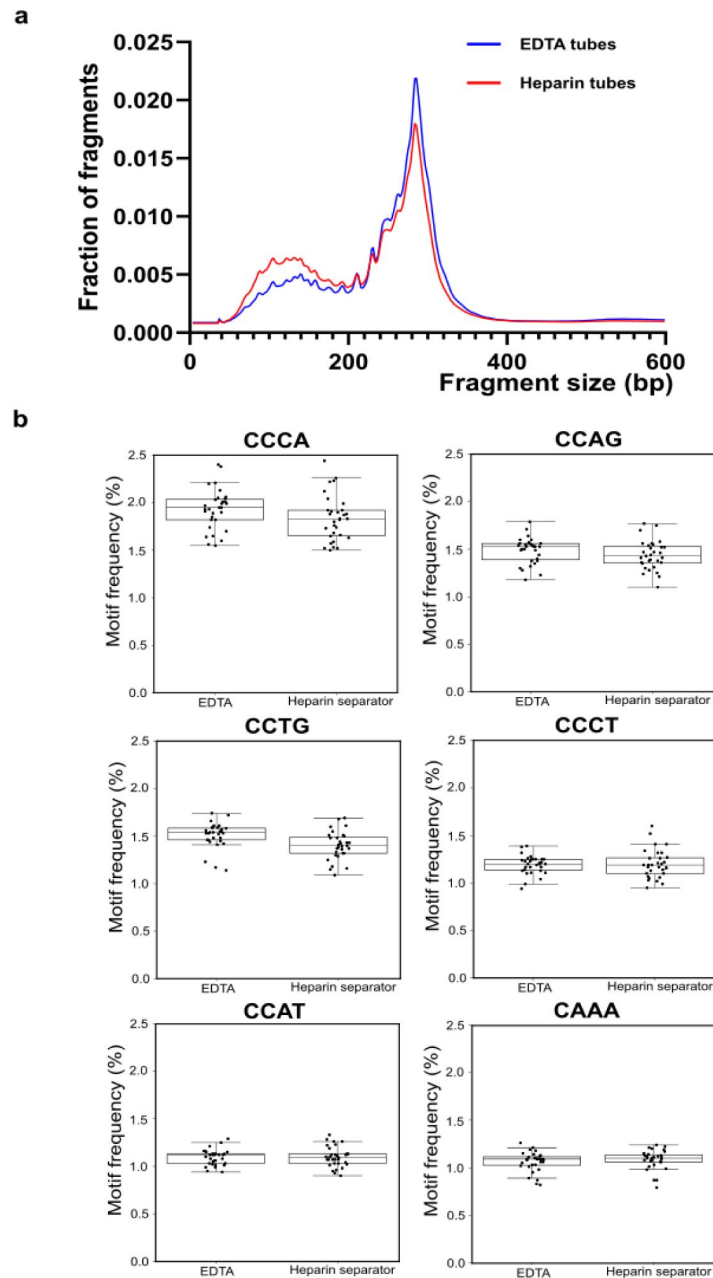


Supplementary Figure 4.

Methylation correlation across Hospital Cohort cases

Heatmaps of methylation beta values in paired samples collected from 11 virus-positive patients in the Hospital Cohort. One additional case is shown in [Figure 4](#).





Supplementary Figure 5.

Fragmentomic patterns of the Hospital Cohort

a. Average cfDNA fragment size distribution collected in EDTA tubes and heparin separators. b. Average end motif frequency comparison between EDTA tubes and heparin separators, highlighting the top six motifs: CCCA, CCAG, CCTG, CCCT, CCAT, and CAAA.

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Author information

Spencer C Ding

Department of Pathology, School of Medicine, Stanford University, Stanford, United States

Jingru Yu

Department of Pathology, School of Medicine, Stanford University, Stanford, United States

Tiepeng Liao

Department of Pathology, School of Medicine, Stanford University, Stanford, United States

Lauren S Ahmann

Department of Molecular and Medical Genetics, Oregon Health & Science University, Portland, United States

Yvette Y Yao

School of Medicine, University of British Columbia, Vancouver, Canada

Chandler Ho

Clinical Laboratories, Stanford Health Care, Stanford, United States

Linlin Wang

Department of Laboratory Medicine, School of Medicine, University of California San Francisco, San Francisco, United States

Benjamin A Pinsky

Department of Pathology, School of Medicine, Stanford University, Stanford, United States, Department of Medicine, Division of Infectious Diseases and Geographic Medicine, School of Medicine, Stanford University, Stanford, United States
ORCID iD: [0000-0001-8751-4810](https://orcid.org/0000-0001-8751-4810)

Wei Gu

Department of Pathology, School of Medicine, Stanford University, Stanford, United States

ORCID iD: [0000-0002-2480-6645](https://orcid.org/0000-0002-2480-6645)

For correspondence: weigu@stanford.edu

Editors

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Gang Peng

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Fred Hutch Cancer Center, Seattle, United States of America

Reviewer #1 (Public review):

Summary:

The manuscript "Adapting Clinical Chemistry Plasma as a Source for Liquid Biopsies" addresses a timely and practical question: whether residual plasma from heparin separator tubes can serve as a source of cfDNA for molecular profiling. This idea is attractive, since such samples are routinely generated in clinical chemistry labs and would represent a vast and accessible resource for liquid biopsy applications. The preliminary results are encouraging, but in its current form, the study feels incomplete and requires additional work.

My major concerns/suggestions are as follows:

(1) Context and literature

The introduction provides only limited background on prior attempts to use heparinized plasma for cfDNA work. It is well known that heparin can inhibit PCR and sequencing library preparation, which has historically discouraged its use. The authors should summarize the relevant literature more comprehensively and explain clearly why this approach has not been widely adopted until now, and how their work differs from or overcomes these earlier challenges.

(2) Genome-wide coverage

The analyses focus on correlations in methylation patterns and fragmentation metrics, but there is no evaluation of sequencing coverage across the genome. For both WGS and WMS, it would be important to demonstrate whether cfDNA from heparin plasma provides unbiased coverage, or whether certain genomic regions are systematically under-represented. A comparison against coverage profiles from cell-derived DNA (e.g., PBMC genomic DNA) would help to put the results in context and assess whether the material is suitable for whole-genome applications.

(3) Viral detection sensitivity

The study shows strong concordance in viral detection between EDTA and heparin samples, but the sensitivity analysis is lacking. For clinical relevance, it is critical to demonstrate how well heparin-derived plasma performs in low viral load cases. A quantitative comparison of viral read counts and genome coverage across tube types would strengthen the conclusions.

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Reviewer #2 (Public review):**Summary:**

The authors propose that leftover heparin plasma can serve as a source for cfDNA extraction, which could then be used for downstream genomic analyses such as methylation profiling, CNV detection, metagenomics, and fragmentomics. While the study is potentially of interest, several major limitations reduce its impact; for example, the study does not adequately address key methodological concerns, particularly cfDNA degradation, sequencing depth limitations, statistical rigor, and the breadth of relevant applications.

Strengths:

The paper provides a cheap method to extract cfDNA, which has broad application if the method is solid.

Weaknesses:

- (1) The introduction lacks a sufficient review of prior work. The authors do not adequately summarize existing studies on cfDNA extraction, particularly those comparing heparin plasma and EDTA plasma. This omission weakens the rationale for their study and overlooks important context.
- (2) The evaluation of cfDNA degradation from heparin plasma is incomplete. The authors did not compare cfDNA integrity with that extracted from EDTA plasma under realistic sample handling conditions. Their analysis (lines 90-93) focuses only on immediate extraction, which is not representative of clinical workflows where delays are common. This is in direct conflict with findings from Barra et al. (2025, LabMed), who showed that cfDNA from heparin plasma is substantially more degraded than that from EDTA plasma. A systematic comparison of cfDNA yields and fragment sizes under delayed extraction conditions would be necessary to validate the feasibility of their proposed approach.
- (3) The comparison of methylation profiles suffers from the same limitation. The authors do not account for cfDNA degradation and the resulting reduced input material, which in turn affects sequencing depth and data quality. As shown by Barra et al., quantifying cfDNA yield and displaying these data in a figure would strengthen the analysis. Moreover, the statistical method applied is inappropriate: the authors use Pearson correlation when Spearman correlation would be more robust to outliers and thus more suitable for methylation and other genomic comparisons.
- (4) The CNV analysis also raises concerns. With low-coverage WGS (~5X) from heparin-derived cfDNA, only large CNVs (>100 kb) are reliably detectable. The authors used a 500 kb bin size for CNV calling, but they did not acknowledge this as a limitation. Evaluating CNV detection at multiple bin sizes (e.g., 1 kb, 10 kb, 50 kb, 100 kb, 250 kb) would provide a more complete picture. In addition, Figure 3 presents CNV results from only one sample, which risks bias. Similar bias would exist for illustrations of CNVs from other samples in the supplementary figures provided by the authors. Again, Spearman correlation should be applied in Figure 3c, where clear outliers are visible.
- (5) It is important to point out that depth-based CNV calling is just one of the CNV calling methods. Other CNV calling software using SNVs, pair-reads, split-reads, and coverage depth for calling CNV, such as the software Conserting, would be severely affected by the low-quality WGS data. The authors need to evaluate at least two different software with specific algorithms for CNV calling based on current WGS data.
- (6) The authors omit an important application of cfDNA: somatic mutation detection. Degraded cfDNA and reduced sequencing depth could substantially impact SNV calling

accuracy in terms of both recall and precision. Assessing this aspect with their current dataset would provide a more comprehensive evaluation of heparin plasma-derived cfDNA for genomic analyses.

<https://doi.org/10.7554/eLife.108708.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

The manuscript "Adapting Clinical Chemistry Plasma as a Source for Liquid Biopsies" addresses a timely and practical question: whether residual plasma from heparin separator tubes can serve as a source of cfDNA for molecular profiling. This idea is attractive, since such samples are routinely generated in clinical chemistry labs and would represent a vast and accessible resource for liquid biopsy applications. The preliminary results are encouraging, but in its current form, the study feels incomplete and requires additional work.

We thank the reviewer for the encouragement and for recognizing the potential of clinical chemistry plasma as an accessible source for cfDNA-based analyses. We look forward to addressing the gaps described below.

My major concerns/suggestions are as follows:

(1) Context and literature

The introduction provides only limited background on prior attempts to use heparinized plasma for cfDNA work. It is well known that heparin can inhibit PCR and sequencing library preparation, which has historically discouraged its use. The authors should summarize the relevant literature more comprehensively and explain clearly why this approach has not been widely adopted until now, and how their work differs from or overcomes these earlier challenges.

We thank the reviewer for their valuable comments and agree that the review of prior work needs to be more thorough, with the gaps clearly identified. In the revised manuscript, we will expand the introduction to include a more comprehensive summary of prior studies. Some of the material was in the Discussion, but we will move it to the introduction in the revision. In general, we will comment briefly here about the novelty of this work and the previous gap in the literature:

(1) Previous pre-analytical studies use DNA fluorometry and qPCR, which cannot distinguish between genomic DNA contamination (from cells) and cfDNA. In contrast, our study uses adapter-based NGS with DNA spike-ins, which can exclude genomic DNA contamination and enable precise quantification of cfDNA input and measurement of their lengths. In Figure 5b-c, we demonstrate that we were able to match our paired sample results only under the measurements of our NGS study, not in previous attempts. Note the current Fig. 5 captions b&c should be swapped and will be corrected in the revision.

(2) As the reviewer has astutely mentioned, heparin is a well-recognized inhibitor of PCR, and heparinized specimens are historically contraindicated for molecular testing. However, most modern cfDNA assays now use NGS, which includes multiple purification steps before PCR amplification, minimizing the impact of heparin interference.

(3) Previous clinical chemistry tests used serum tubes, which are known to generate background gDNA during clotting and are therefore unsuitable for cfDNA-based analyses. In recent years, modern hospital chemistry laboratories, especially those supporting emergency departments, have gradually transitioned to heparin separator tubes for faster turnaround. Hence, residual plasma from heparin separator tubes is a more recent option, one that was not widely available when key pre-analytical studies on cfDNA were performed.

(2) Genome-wide coverage

The analyses focus on correlations in methylation patterns and fragmentation metrics, but there is no evaluation of sequencing coverage across the genome. For both WGS and WMS, it would be important to demonstrate whether cfDNA from heparin plasma provides unbiased coverage, or whether certain genomic regions are systematically under-represented. A comparison against coverage profiles from cell-derived DNA (e.g., PBMC genomic DNA) would help to put the results in context and assess whether the material is suitable for whole-genome applications.

Thank you for the insightful comment. We agree that evaluating sequencing coverage across the genome is important for assessing the suitability of cfDNA from heparin separators. In response, we are performing additional, in-depth runs to compare genome-wide coverage profiles in the Hospital Cohort. The results of these analyses will be included in the revised version of the manuscript.

(3) Viral detection sensitivity

The study shows strong concordance in viral detection between EDTA and heparin samples, but the sensitivity analysis is lacking. For clinical relevance, it is critical to demonstrate how well heparin-derived plasma performs in low viral load cases. A quantitative comparison of viral read counts and genome coverage across tube types would strengthen the conclusions.

We agree that evaluating analytical sensitivity in cases with low viral loads is important for understanding clinical performance. To address this point, we plan to include additional paired cases with viral loads below 1,000 IU/mL and examine the correlation of viral read counts between EDTA and heparin separators in this subset.

Reviewer #2 (Public review):

Summary:

The authors propose that leftover heparin plasma can serve as a source for cfDNA extraction, which could then be used for downstream genomic analyses such as methylation profiling, CNV detection, metagenomics, and fragmentomics. While the study is potentially of interest, several major limitations reduce its impact; for example, the study does not adequately address key methodological concerns, particularly cfDNA degradation, sequencing depth limitations, statistical rigor, and the breadth of relevant applications.

We thank the reviewer for the insightful comments and will work to clarify and address the mentioned issues. We do not find the residual plasma from the heparin separator to be a replacement for gold standard methods. Instead, we take it as a practical and complementary resource that may help broaden the accessibility of samples. Comparable cfDNA metrics highlight its potential to serve as an additional source for biobanking and research applications.

Strengths:

The paper provides a cheap method to extract cfDNA, which has broad application if the method is solid.

We thank the reviewer for the encouraging comment. While cost-effectiveness is a practical advantage, we believe the greater strength of this approach lies in the accessibility of sampling. Residual plasma from routine clinical tests offers an opportunity to include patients or time points that would otherwise be difficult to capture, such as those with severe illness or those sampled before treatment.

Weaknesses:

(1) The introduction lacks a sufficient review of prior work. The authors do not adequately summarize existing studies on cfDNA extraction, particularly those comparing heparin plasma and EDTA plasma. This omission weakens the rationale for their study and overlooks important context.

We thank both reviewers for this comment. See above under Reviewer 1's responses for our provisional perspective on the background literature and gap. We will expand the Introduction to provide a more comprehensive summary of prior studies.

(2) The evaluation of cfDNA degradation from heparin plasma is incomplete. The authors did not compare cfDNA integrity with that extracted from EDTA plasma under realistic sample handling conditions. Their analysis (lines 90-93) focuses only on immediate extraction, which is not representative of clinical workflows where delays are common. This is in direct conflict with findings from Barra et al. (2025, LabMed), who showed that cfDNA from heparin plasma is substantially more degraded than that from EDTA plasma. A systematic comparison of cfDNA yields and fragment sizes under delayed extraction conditions would be necessary to validate the feasibility of their proposed approach.

We appreciate this thoughtful comment, which highlights reasonable concerns about cfDNA degradation in heparin. We would like to clarify that the Hospital Cohort, which only used leftover plasma in the clinical lab, was designed to reflect real-world clinical workflows, where unavoidable delays before plasma processing are already incorporated. In the Healthy Cohort, a subset of samples is also processed after controlled delays, as shown in Supplementary Figure 2.

Regarding the differing results in Barra et al. (2025, LabMed), where heparin tubes showed 85% cfDNA degradation, it is important to note that samples were incubated at 37 °C for 24 hours. We anticipate that endogenous nuclease would be active under 37 °C and would cause cfDNA degradation. However, this condition differs markedly from the relevant clinical workflows we describe here. In the routine hospital settings, blood samples are typically kept at room temperature for up to 60 minutes during transport and waiting. The outpatient setting can be more variable, but samples here are supposed to be refrigerated during transportation. They are then processed in high-throughput, fully automated systems that comply with nationally standardized quality regulations in the United States (CLIA). The resultant plasma will be physically separated from cellular components because of the gel in the heparin separators. The processed tubes are subsequently transferred to refrigerated storage at 4 °C. Under these conditions, samples do not experience prolonged exposure to elevated temperatures such as 37 °C, and refrigeration usually occurs within two hours of collection. We will incorporate these details in the revised manuscript.

Also, as we mentioned in our reply to Reviewer 1, Barra et al. used qPCR like most cfDNA pre-analytical studies, but qPCR is not a perfect DNA quantification method for NGS-based downstream analyses because it measures both cfDNA and contaminating genomic DNA. The latter can be excluded by most NGS assays. By using constant spike-in internal controls, our approach directly quantifies the amount of sequenceable cfDNA, providing a more accurate estimate of input DNA (Figure 5c). In one possible future experiment, the same sample in the Healthy Cohort can be delayed by 1-2 hours prior to processing (centrifugation and refrigeration) and kept at room temperature rather than 4 °C to mimic real-world delays. Outputs would be cfDNA yields and fragment sizes, and we would use constant spike-ins to quantify the amount of sequenceable DNA.

(3) The comparison of methylation profiles suffers from the same limitation. The authors do not account for cfDNA degradation and the resulting reduced input material, which in turn affects sequencing depth and data quality. As shown by Barra et al., quantifying cfDNA yield and displaying these data in a figure would strengthen the analysis. Moreover, the statistical method applied is inappropriate: the authors use Pearson correlation when Spearman correlation would be more robust to outliers and thus more suitable for methylation and other genomic comparisons.

We appreciate the reasonable concerns regarding cfDNA degradation and agree that the methylation profile is not an adequate metric for degradation. To evaluate for degradation, we will focus on NGS-derived length profiles (WGS data) and constant spike-in DNA. We appreciate the reviewer's suggestion to use the Spearman correlation, and this will be incorporated.

(4) The CNV analysis also raises concerns. With low-coverage WGS (~5X) from heparin-derived cfDNA, only large CNVs (>100 kb) are reliably detectable. The authors used a 500 kb bin size for CNV calling, but they did not acknowledge this as a limitation. Evaluating CNV detection at multiple bin sizes (e.g., 1 kb, 10 kb, 50 kb, 100 kb, 250 kb) would provide a more complete picture. In addition, Figure 3 presents CNV results from only one sample, which risks bias. Similar bias would exist for illustrations of CNVs from other samples in the supplementary figures provided by the authors. Again, Spearman correlation should be applied in Figure 3c, where clear outliers are visible.

We appreciate the reviewer's constructive comments regarding the CNV analysis. We agree that the use of low-coverage WGS (~5×) limits the reliable detection of small CNVs, and we will acknowledge this as a limitation in the revised manuscript. To address this point, we will perform additional analyses using 50kb as bin sizes. To reduce potential bias from single-sample representation, we will show the aggregated CNV plots for all CNA-positive cases along with their log₂ copy ratio correlations, and Spearman's correlation will be applied as suggested.

(5) It is important to point out that depth-based CNV calling is just one of the CNV calling methods. Other CNV calling software using SNVs, pair-reads, split-reads, and coverage depth for calling CNV, such as the software Conserting, would be severely affected by the low-quality WGS data. The authors need to evaluate at least two different software with specific algorithms for CNV calling based on current WGS data.

Thank you for this suggestion. We will evaluate CNV profiles using alternative informatics methods.

(6) The authors omit an important application of cfDNA: somatic mutation detection. Degraded cfDNA and reduced sequencing depth could substantially impact SNV calling

accuracy in terms of both recall and precision. Assessing this aspect with their current dataset would provide a more comprehensive evaluation of heparin plasma-derived cfDNA for genomic analyses.

We thank the reviewer for emphasizing SNVs as an important application of cfDNA. We agree that the limited volume of residual plasma is a constraint. Routine chemistry tests leave ~1–2 mL of plasma, and this limited volume places an upper limit on performing SNV analysis. We will expand the discussion of this limitation in the paper. Our approach is not intended to replace specialized tubes for large-volume cfDNA collection but rather to complement them by enabling the use of residual material.

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